



```
...     print("3-DES provides better security than DES,")
...     print("but it requires more processing time.")
...
```

=====

### 3-DES (TRIPLE DES) ALGORITHM

=====

```
Enter plaintext: Hello world
Enter Key 1 (8 characters): 12345678
Enter Key 2 (8 characters): abcdefgh
Enter Key 3 (8 characters): qwertyui
```

#### ----- RESULT -----

```
Original Text      : Hello world
Key 1              : 12345678
Key 2              : abcdefgh
Key 3              : qwertyui
Encrypted Text     : 2d7f5ca5f248d62ca672fe63fc75e637
Decrypted Text     : Hello world
```

```
Encryption Time    : 0.0036393998889252543 seconds
Decryption Time    : 0.002585100010037422 seconds
```

```
Verification       : SUCCESS
Original plaintext  successfully recovered.
```

#### ----- SECURITY ANALYSIS -----

1. 3-DES applies DES three times.
2. It provides stronger security than single DES.
3. Three different keys are used.
4. 3-DES is slower than DES.
5. 3-DES requires more processing because three DES operations are performed.
6. AES is preferred for modern applications.

#### ----- CONCLUSION -----

```
3-DES provides better security than DES,
but it requires more processing time.
```